UNIT – 5

ASSESSMENT – 1

SOLUTION

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1. The system uses two recommendation approaches.

Content-Based Filtering

Recommends items similar to items previously liked by the user.

For example:

User likes Python Course

↓

System identifies similar courses

↓

Recommends Data Science Course

Collaborative Filtering

Finds users with similar preferences and recommends items liked by similar users.

User A and User B have similar preferences

↓

User B likes Item X

↓

Recommend Item X to User A

Hybrid Recommendation

The final score can be calculated as:

where:

α = weight given to content-based recommendation

Intelligent Agent

The agent monitors:

- User clicks
- Ratings
- Purchases
- Likes/dislikes
- Feedback

and updates recommendations accordingly.

2. The system uses PySpark RDDs to process data from multiple sources.

Data Sources

Sensors

↓

Satellite Data

↓

Social Media

↓

PySpark RDD

↓

Data Fusion

↓

Disaster Detection

↓

Intelligent Agent

↓

Alert

Data Fusion

Data from different sources is combined using a common location and disaster type.

For example:

Sensor → High water level

Satellite → Flooded area

Social media → "Flood reported"

Together these increase the confidence of flood detection.

Intelligent Agent

The agent:

- Monitors incoming information.
- Calculates disaster confidence.
- Generates warnings.
- Recommends response actions.
- Updates its decision using new observations.

3. A transaction can be considered suspicious when it has abnormal characteristics such as:

Very high amount

+

Unusual location

+

High transaction frequency

↓

Possible Fraud

PySpark RDDs distribute transaction processing across multiple workers.

The intelligent agent monitors suspicious transactions and generates alerts.

4. The system processes inventory and demand data using RDDs.

For example:

Demand data

↓

RDD processing

↓

Inventory analysis

↓

Stock shortage detection

↓

Supply Chain Agent

↓

Restock recommendation

5. The system analyzes student learning data using distributed RDD processing.

Example:

Student Activity

↓

PySpark RDD

↓

Performance Analysis

↓

Weak Topic Detection

↓

Learning Agent

↓

Personalized Recommendation

Intelligent Learning Agent

The agent monitors:

- Scores
- Learning time
- Completed lessons
- Quiz performance
- Feedback

Based on these observations, it recommends appropriate learning material.